

LAABH

Large-scale Infrastructure Analytics & Bottleneck Handling
README / Project Documentation

1. Project Overview

LAABH is an intelligent infrastructure project analytics and decision-support platform designed to monitor large-scale projects, identify bottlenecks, analyze risks, and explore possible interventions. It combines a React dashboard, FastAPI backend, project data, machine-learning models, and interactive what-if analysis.

2. Key Features

Project Dashboard: Project details, costs, physical and financial progress, status, risk, recoverability, time overrun, and cost overrun.

Project Diagnostics: Statutory approvals, execution velocity, financial flow, contractor disbursement, systems integration, and commissioning analysis.

Risk Analysis: Risk score, recoverability score/band, time and cost overrun, execution velocity, and progress indicators.

What-If Simulator: Scenario analysis for completion time, risk, recoverability, intervention cost, and economic value.

Project Registry: Infrastructure project identifiers, sectors, ministries, agencies, states, dates, costs, progress, diagnostics, and milestones.

3. Technology Stack

Frontend: React, Vite, JavaScript, Recharts, Lucide React, HTML, CSS.

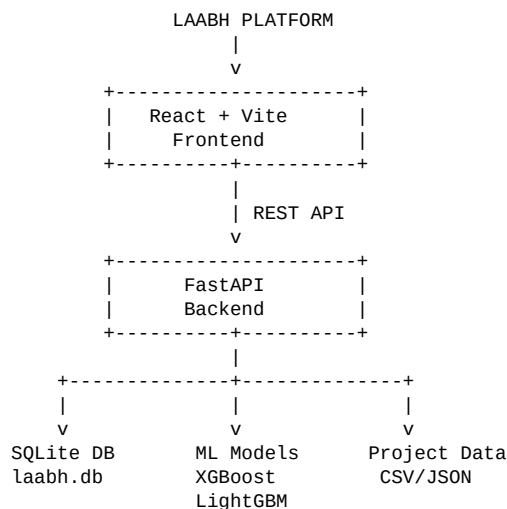
Backend: Python, FastAPI, Uvicorn.

Machine Learning: Scikit-learn, XGBoost, LightGBM, SHAP, Joblib, NumPy, Pandas.

Database: SQLite.

Authentication: JWT and bcrypt.

4. System Architecture



5. Project Structure

```
final-sih/
+-- backend/
|   +-- app/
|   +-- laabh.db
|   +-- requirements.txt
+-- data/
+-- models/
+-- src/
+-- scripts/
+-- tests/
+-- index.html
+-- package.json
+-- package-lock.json
+-- vite.config.js
```

```
+-- styles.css
+-- .gitignore
+-- README.md
```

6. Local Installation

```
git clone https://github.com/YOUR_USERNAME/final-sih.git
cd final-sih

# Backend
cd backend
python -m pip install -r requirements.txt
cd ..
python -m uvicorn backend.app.main:app --reload --port 8000

# Frontend (new terminal)
npm install
npm run dev
```

Backend: <http://127.0.0.1:8000/docs>

Frontend: <http://localhost:5173>

7. Production Build

```
npm run build

Production output: dist/
```

8. Railway Deployment

Deploy the project as two Railway services from the same GitHub repository.

Backend: Root Directory: /backend | Build: pip install -r requirements.txt | Start: uvicorn app.main:app --host 0.0.0.0 --port \$PORT

Frontend: Root Directory: / | Build: npm install && npm run build | Start: npx serve -s dist -l \$PORT

VITE_API_URL=<https://your-backend.up.railway.app>

9. Database and Machine Learning

The current application uses backend/laabh.db as its SQLite database. Saved model artifacts are stored in models/, including XGBoost, LightGBM, calibrated models, and feature-column information.

10. Testing

```
npm run build
python -m uvicorn backend.app.main:app --reload --port 8000
Open: http://127.0.0.1:8000/docs
```

11. Use Cases

Monitor large infrastructure projects; identify bottlenecks; analyze project progress and execution velocity; analyze risk and recoverability; compare financial and physical progress; explore intervention scenarios; support infrastructure project decision-making; provide a centralized analytics dashboard.

12. Future Scope

Real-time government project data integration, automated data ingestion, advanced predictive models, improved forecasting, geographic visualization, automated risk alerts, role-based access control, cloud database integration, model monitoring/retraining, additional infrastructure datasets, and advanced policy simulation.

13. Project Information

Project: LAABH

Repository: final-sih

Architecture: React + FastAPI + Machine Learning

Deployment: Railway

Developed as a student project focused on applying data analytics, machine learning, and modern web technologies to

infrastructure project monitoring and decision support.